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In Mathematics (9MA0)

Paper 01 Pure Mathematics

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General comments

Overall, the 9MA0/01 paper provided plenty of challenge for the full cohort of learners.

This paper helps learners get off to a good start and many learners performed well on the early questions. Many fully correct responses were also seen on Questions 6, 8 and 9 with good attempts at all questions on the paper. It was rare to see a completely blank response to a question. In contrast, the parts that learners found particularly challenging were Q7(a), Q10(c), Q11(b)/(c), Q12(b), Q13(a)/(b), Q14(b) and Q15

Many learners scored well in later parts when responses for earlier parts were not correct. This discriminated between the responses well, with some showing particularly impressive solutions to the problems.

The presentation of solutions was generally good, although a number of responses still do not show all the steps in the solution or lack labelling or little indication as to what is being carried out. This is particularly an issue when the answer is incorrect but there is no method visible to determine whether the learner had made a slip in their method or not. Equally, it can be an issue when there are many attempts, with a lot of incorrect approaches and no indication as to what should be marked. Learners should continue to be mindful of presenting their solutions as clearly as possible.

It was clear on questions such as Question 10 where incorrect calculus was then followed by a correct value that learners are using the functions on their calculator to support their working. Whilst this is useful to check answers and help to maintain accuracy, learners should then make sure that their algebraic expressions are correct as full marks is unlikely to be scored from incorrect working. This use of the calculator, however, did at least ensure that the next part

of the question was started with the correct answer from the previous part. In cases where a quadratic equation needed solving, it was more common to see just the answer stated, which was acceptable provided there was not the calculator warning not to do so; incorrectly putting the values into the calculator, however, can cost both the method and accuracy mark in some cases.

Comments on individual questions

Question 1

This short question on graph transformations was a nice introduction to the paper with many responses scoring well. The question assessed learners understanding of the impact of function transformations on a given point $P(-4, 5)$. Incorrect responses were seen but most learners attempted to answer both parts of the question.

Most responses were given without working but, where it was shown, the most common approach was by coordinate mapping with separate working for each coordinate. A few responses left a calculation rather than value as their final answer e.g. $\frac{-4}{2}$.

Part (a)

This part of the question focused on the impact of transformations involving a stretch (horizontal or vertical). Learners were required to give the coordinates of the point P when $y = f(x)$ was mapped to $y = 4f(2x)$. Many correct responses were seen. Errors were more commonly seen when identifying the x coordinate which should be mapped to -2 . Common incorrect answers included $x = -8$, $x = -4$ and $x = -16$. The y coordinate should be mapped to 20 and incorrect answers included $y = 10$, $y = 5$, $y = 1.25$ and $y = 2.5$.

Part (b)

This part of the question considered the impact of translation (horizontal or vertical) on the coordinates of point P . Learners were required to give the coordinates of the point P when $y = f(x)$ was mapped to $y = f(x+3) - 2$. Overall, this was well-attempted with many correct answers seen. A common

source of error was transforming in the wrong direction, particularly among learners who had demonstrated a similar misconception in part (a). This resulted in an x coordinate of -1 or, less commonly, a y coordinate of 7 .

Question 2

This question looked at how to find a root of an equation by using iteration. The first part investigated how to show that the function had a root that lay within a particular interval and the second part required the learner to find an approximation of the root to 4 decimal places, using a given iterative formula. Overall, part (b) of this question was answered well. However, the accuracy mark in part (a) required learners to state that the function is continuous. Most learners were able to attempt both parts of the question. It was generally accessible for the majority with many giving clear, concise answers.

Part (a)

In part (a), most learners correctly substituted the values of 1.1 and 1.2 into the function, obtaining the required values to the correct accuracy and identifying the change in sign. Many learners went on to achieve the conclusion mark by correctly stating that a root exists within the interval. The most common reason for not gaining marks in this part was the omission of the requirement that the function is continuous. Although most learners recognised that a change in sign was present, many concluded only that the root existed because the signs were different, without referencing continuity. Some answers attempted to include continuity but demonstrated a misunderstanding of the concept, with statements such as “the interval is continuous” rather than referring to the function. Learners should ensure that their final conclusion includes all necessary components: the function is continuous, there is a change in sign, and therefore a root exists in the given interval.

Part (b)

Part (b) was also answered very well overall, with the majority of learners successfully applying the iteration formula and obtaining the required values. Most learners correctly completed the first iteration and achieved the answer of 1.1542. Many learners demonstrated a good understanding of the iterative process, with some continuing the iterations until the values remained constant to the required number of decimal places. Some learners obtained the correct iteration but did not round their answer correctly to 4 decimal places, instead giving answers to 3 decimal places or providing a value which did not round to the required accuracy.

In the final part of the iteration, most learners who successfully completed the earlier section were able to continue and obtain the correct limiting value. The main misconception was that some learners stopped after finding the next term rather than continuing the iteration until the required accuracy had been reached. Other responses seen showed every iteration until the decimal places stopped changing; although this was unnecessary, it demonstrated a secure understanding of convergence.

Question 3

This question on differentiation and coordinate geometry was a very accessible question that was answered well by the vast majority of learners. It assessed learners' ability to differentiate a polynomial function before using the derivative to determine the equation of the tangent to the curve at a given value of x .

Part (a)

Most responses seen answered the question well by correctly differentiating the polynomial and demonstrating a secure understanding of polynomial differentiation. The derivative was often written down directly without additional

working. Where marks weren't awarded, this was usually due to minor slips, such as incorrect differentiation of the $-2x^2$ term or coefficient errors. Very occasionally, learners attempted to integrate rather than differentiate.

Part (b)

This part was also answered well, with many learners successfully finding the equation of the tangent. Better responses correctly evaluated both the function and its derivative at the given value of x to determine the point of contact and the gradient before using either the point-gradient form or the $y = mx + c$ form of the equation of a straight line. Neither method appeared to offer a particular advantage, and both were used successfully.

Where responses were less successful, the most common misconception was to find the equation of the normal rather than the tangent by using the negative reciprocal of the gradient. A small number incorrectly used the second derivative as the gradient of the tangent. Other errors typically arose from arithmetic slips when evaluating the coordinates of the point or the gradient, or from algebraic or arithmetical errors when rearranging the final equation. A common incorrect final answer arising from otherwise correct working was $y = 9x - 21$. Occasionally, learners substituted into the incorrect expression when finding the y coordinate or used an incorrect value of x .

Centres may wish to continue encouraging learners to show key stages of their substitutions and calculations, as this can help reduce arithmetic errors and ensure that method marks are secured even where numerical slips occur. Overall, this question proved accessible to the majority of learners and demonstrated a strong understanding of differentiation and the relationship between a derivative and the gradient of a tangent.

Question 4

This question required the learners to work with functions, giving their range, finding the value of a composite function and finding the inverse function. Most responses were good or fully successful attempts.

Part (a)

Part (a) was generally well answered by learners identifying -2 as the value required for the range of the function. Responses which did not get the mark included $f(x) > -2$, $y > -2$ or $x \geq -2$.

Part (b)

In part b most learners successfully found $f(2)$ followed by $g(10)$ giving the exact answer $\frac{49}{6}$, though some gave a rounded answer of 8.17 or 8.2 and did not gained the final mark. Of those who attempted to find an expression for $gf(x)$ first then substitute 2, some made slips in their algebraic manipulation.

Part (c)

This was attempted and showed that learners have a good understanding of the method to find the inverse function.

Miscopying the question with e.g. a sign error was often seen, as were sign slips in algebraic manipulation, particularly if an attempt was made to cross multiply and expand in one step. A few learners did not isolate x in the term

$$yx - 5x - 4y = -1 \text{ and proceeded to } x = \frac{-1}{y - 5x - 4}$$

Of those who used the method of starting with $y = \frac{5x-1}{x-4}$ arriving at $x = \frac{4y-1}{y-5}$, some did not go on to express the final answer in terms of x .

Many learners thought they had to give the domain though it was not required as the question asked for the expression of $g^{-1}(x)$ not the domain of g^{-1} .

Question 5

This question tested understanding of parametric equations within a context. It was evident that learners seemed more confident with the concept of parametric differentiation which was not required for this particular question and as a result a number of responses demonstrated an incorrect approach to the problem.

Part (a)

Learners were asked to find the maximum height of the helicopter above the ground. This was answered extremely well with correct responses given as either 10 or 10 m.

Part (b)

Learners were asked to determine the horizontal distance when the helicopter was 5m above the ground for the second time.

The most common method used was to substitute 5 into $y = 10\sin\left(\frac{t^2}{4}\right)$ and rearrange to find at least one value for t . Learners would then likely substitute their value for t into $x = 100\sqrt{t}$ to find the distance. Learners were allowed to do their workings in either degrees or radians, so answers of 180 m or 495 m were

accepted. There was a fairly even split between learners using radians and those using degrees.

The most common errors seen included omitting to find the second value for t , with learners often mistakenly substituting values found from $\frac{t^2}{4} = 30$ or $\frac{t^2}{4} = \frac{\pi}{6}$.

Other errors included incorrect processing from their $5 = 10\sin\left(\frac{t^2}{4}\right)$ when finding a value for t then x . Notable processing errors included learners confusing t^2 , $\sqrt{t}$ and t

Alternatively, learners attempted to first find a cartesian equation, by eliminating t to find an equation of the form $y = C\sin(Dx^4)$. The most commonly seen

intermediate step was to obtain an expression for $y = 10\sin\left(\frac{\left(\frac{x}{100}\right)^4}{4}\right)$.

Learners would then typically set their expression for y equal to 5 and then find a value for x . This approach proved more problematic for learners with processing errors being commonplace.

Question 6

This question assessed learners' knowledge of integrating an algebraic fraction and algebraic manipulation involving the natural log function. Learners were able to access both parts of this question but many did not gain the accuracy mark in both parts.

Part (a)

In part (a), most learners recognised that they should integrate to the form $A \ln(7x + 1)$. Common reasons for not gaining the accuracy mark included: having an incorrect coefficient after integration e.g. $21 \ln(7x + 1)$ or $\frac{7}{3} \ln(7x + 1)$; missing the constant of integration, $+c$; missing brackets or modulus signs around the $7x + 1$ in their final answer; and less commonly was an attempt to integrate the function as a polynomial. This would generally result in some equation of the form $k \times (7x + 1)^{\pm 2} + c$, which did not score any marks. On even rarer occasions, answers incorrectly manipulated the integrand as $\int \frac{3}{7x+1} dx = \int \left(\frac{3}{7}x + 3 \right) dx$ and integrated as a polynomial.

Part (b)

In part (b), many learners demonstrated clarity of working leading to a correct final answer or a “correct final answer” following through on their part (a) answer. Stronger responses included a working line showing the substitution of p and 2 into their answer from part (a), subtracting the correct way around and equating to the required fraction. Some learners were able to recognise that the whole equation could be multiplied by $\frac{7}{3}$ and therefore went straight to $= 2$ on the RHS, avoiding later fraction manipulation. Assumptions that substituting 0 into the integral would produce an answer of 0 were relatively uncommon, as was the accidental inclusion of $+c$ in the exact integral.

Learners who applied a law of logarithms to combine their terms and obtain $\ln\left(\frac{7p+1}{15}\right) = 2$ were much more likely to be successful in achieving full marks for this part of the question.

Where responses did not gain the final two marks, it was usually for difficulty to convert from an equation involving $\ln$ to an equation involving e correctly. Weaker responses did not combine terms involving $\ln$ and therefore found the manipulation of the equation to the correct result more of a challenge.

Occasionally weaker responses had arithmetical errors when processing the coefficients, but most were aware of the need for an exact answer, and it was rare to see $\ln 15$ processed as a decimal. A less common, and less successful approach, was to attempt rearrangement without combining the log terms, $\ln(7p+1) = \ln(15) + 2$ was written as $7p + 1 = e^{\ln(15) + 2}$, stronger responses correctly using the laws of indices to proceed to $7p + 1 = 15e^2$. Some weaker responses did not gain the final two marks as they proceeded incorrectly to give $7p + 1 = 15 + e^2$.

Question 7

This question on radian measure and the Newton-Raphson method was one of the most demanding questions on the paper. Learners found the segment areas manipulation and absence of numerical values in (a) to be quite challenging, so it was not uncommon for attempts to be abandoned, leaving part (b) unattempted. However, other learners did make good progress with this question and some responses saw multiple attempts in part (a) or working algebraically in part (b), which still scored the majority of marks in that part.

Part (a)

Many learners were able to gain the opening marks by forming expressions for the areas of the two segments. Stronger responses recognised that the angle in the second segment was $\pi - \theta$ and formed correct equations linking the two areas. Learners who identified immediately that $\sin(\pi - \theta) = \sin \theta$ generally made more efficient progress than those who chose to expand using compound angle

formulae.

A common source of error was identifying the angle in the second segment, with $2\pi - \theta$, $180 - \theta$ or $\theta - \pi$ frequently seen. Some learners initially worked in degrees before later replacing 180 with π , although this was not always completed successfully. There was also some confusion when interpreting the relationship between the two segments of circle, with some responses treating R_2 as twice R_1 rather than recognising that R_1 was twice R_2 .

Many responses lost accuracy when simplifying expressions involving $\sin(\pi - \theta)$. Whilst some learners correctly replaced this directly with $\sin \theta$, others attempted to expand using compound angle formulae and introduced sign errors, often obtaining $-\sin \theta$. Errors were also seen through omitted brackets or incorrect expansions such as $\sin \pi - \sin \theta$. Careless algebraic manipulation during rearrangement, particularly involving negative signs, also prevented otherwise valid methods from reaching the required result.

Part (b)

Learners who attempted this part generally demonstrated a good understanding of the Newton-Raphson method. Successful responses frequently worked correctly using values of p and q , although valid follow-through marks were available for learners who continued using values obtained from their own work in part (a).

Most learners differentiated successfully, whether using given values or working in terms of p and q . The most common differentiation error was treating the constant term -2π as though it differentiated to -2π rather than 0. Sign errors were occasionally seen when applying the Newton-Raphson iteration.

Many learners who had not completed part (a) did not attempt part (b), despite valid follow-through methods being available. Learners should be encouraged to continue with subsequent parts using their own previous results wherever possible, as method marks remain available even when earlier numerical errors have been made. On this occasion, it was also acceptable to work algebraically and still score two of the three marks.

Question 8

This was a familiar type of question which used the concept of integration as the limit of a sum and then subsequently carrying out integration; in this case integration by parts was required. Many responses scored the majority of the marks with coefficient or sign slips being the most common reasons preventing fully correct solutions.

Part (a)

Learners were asked to express $\lim_{\delta x \rightarrow 0} \sum_{x=0}^3 x e^{2x} \delta x$ as an integral.

This was usually answered well with a fully correct answer including all correct notation. The most common error seen here was omitting dx in their $\int_0^3 x e^{2x} dx$. However, this was allowed to be recovered if seen again in part (b).

Other errors included omitting the limits or swapping the limits. Also seen were answers such as $\int_0^3 x e^{2x} \delta x dx$ or other incorrect variables used. There were other miscopying errors seen and occasionally additional incorrect notation was written.

Part(b)

Learners were then asked to find a value for this integral in the form $Ae^6 + B$

The most common approach was to use integration by parts or an equivalent method such as the DI method. Integration by parts was the most successful method chosen.

Learners were usually able to obtain an expression of the form $\alpha xe^{2x} - \beta \int e^{2x} dx$

and then typically apply integration to $\frac{1}{2}xe^{2x} - \frac{1}{2} \int e^{2x} dx$ and obtain the correct

intermediate step of $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}$ as required. Common errors at this point

included not dealing with coefficients correctly when integrating or having a first term in x^2 . It was also notable that many learners did not write the formula for

integration by parts and also that the expression $\frac{1}{2}xe^{2x} + \frac{1}{2} \int e^{2x} dx$ was common.

Some learners chose the wrong values for their u and v when applying integration by parts, effectively swapping them round, thus making the integration more complicated and making little further progress.

Learners were then asked to apply the limits to their integral, and this mark was dependent on having achieved a required form following integration. Common errors included the assumption that substituting 0, gives a value of 0. Other

common errors included obtaining an answer of $\frac{5}{4}e^6 - \frac{1}{4}$, although this slip was

condoned for the final method mark.

A few learners chose to use integration by substitution, with $u = 2x$, the most common approach. Learners would then use integration by parts on their changed expression. They would usually have some difficulty at this point as their

new integral was no more easy than their previous, and they also may not have changed the limits.

Question 9

This question also proved to be more demanding than the earlier questions on the paper. Learners were first required to prove a trigonometric identity involving an integer constant, k , before using the result from part (a) to solve a trigonometric equation over a given interval. The use of reciprocal trigonometric functions and a less familiar interval added to the demand of the question. A calculator warning was included, and learners were instructed to show all stages of working.

Part (a)

Many learners made a positive start by expanding the brackets and rewriting the reciprocal trigonometric functions in terms of sine and cosine, allowing them to access the first method mark. Most were then able to simplify the expression, with common intermediate forms including $\cot\theta - \tan\theta$, $\frac{\cos^2\theta - \sin^2\theta}{2\sin\theta\cos\theta}$, or the single fraction $\frac{1 - \tan^2\theta}{\tan\theta}$.

Better responses recognised how these expressions could be rewritten using the appropriate double-angle identities, before correctly concluding that $k = 2$. A variety of valid approaches were seen, although remaining in terms of sine and cosine throughout often proved to be the more straightforward method.

Where responses were less successful, the main difficulties arose in selecting and applying the appropriate trigonometric identities once the expression had been simplified. Many learners reached intermediate expressions such as $\cot\theta - \tan\theta$ but did not show that they are able to make the connection to the

required double-angle identities. Common incorrect answers included $k = \frac{1}{2}$, usually resulting from incorrect use of the tangent double-angle identity, and $k = 12$, often caused by errors in the final stages of simplification. A small number attempted to prove the identity by working from both sides or by trying to 'meet in the middle', although these approaches were rarely successful. Occasionally, reciprocal trigonometric functions were recalled incorrectly, or intermediate stages of working were omitted, limiting the credit that could be awarded. Centres may wish to continue developing learners' confidence in selecting appropriate identities and presenting clear, sequential proofs.

Part (b)

Learners who obtained the correct value of k in part (a) generally made good progress through the remainder of the question. However, there were also many examples of learners gaining credit by following through with their own value of k . A number of learners who had not completed part (a) correctly used or inferred that $k = 2$, allowing them to access the majority of the available marks. Use of a calculator would have enabled learners to deduce that $k = 2$ by substituting in a value into both sides of the identity which would have enabled them to proceed with the correct value. Others selected an alternative value of k in order to begin the question and, where this was used consistently, were still able to gain the available method marks, although they could not obtain the correct final solutions. The first mark was also available if the response contained the unknown k . Better responses typically rewrote the equation either as a quadratic in $\cot 2x$ or as an equation involving $\sin 4x$. Both approaches were successful, although the method using $\sin 4x$ often made it easier to identify all four required solutions. Once an appropriate trigonometric equation had been established, many learners demonstrated secure algebraic manipulation and confidently solved for the required values of x .

Where responses were less successful, marks were most commonly not gained through incorrect recall or application of trigonometric identities, omission of the factor of $2x$, or errors when manipulating the equation into a suitable form. Common errors included obtaining $\sin 4x = 0.4$ instead of $\sin 4x = 0.8$ through incorrect use of the double-angle formula or forming equations in $\cot x$ or $\sin 2x$ rather than $\cot 2x$ or $\sin 4x$. Some learners relied on calculator-derived answers without providing sufficient supporting working, despite the instruction that all stages should be shown.

The final stage of identifying every solution within the given interval proved to be the greatest challenge. Many learners found only one or two correct solutions, omitted valid solutions, included values outside the stated interval or introduced invalid solutions such as $x = 0$ not realising that this did not hold true since the original substitution required that $x \neq 90n$ (where n could be 0). Centres may wish to continue emphasising the importance of clear notation and well-structured working alongside the accurate recall and application of trigonometric identities, particularly in proof and multi-stage problem-solving questions.

Question 10

This question tested learners' understanding of calculus, requiring them to use differentiation to find the equation of a normal line to a curve, and find the area between a line and the curve using integration. Most responses showed strong engagement with the question, particularly in parts (a) and (b). Part (c) proved more demanding, with many responses demonstrating difficulties in integrating correctly or in identifying the correct region.

Part (a)

In part (a), almost all learners successfully found the y coordinate. For differentiating the function, the use of the quotient rule was more popular than

the product rule. The most common error was to not apply the chain rule correctly on the components, though some learners made arithmetic errors. Learners often chose to attempt to simplify their resulting expression from either the product or quotient rule and sometimes made algebraic errors.

Learners who did not achieve a correct derivative were generally able to obtain further marks by correctly substituting $x = 2$ into their derivative to obtain a value which was then used correctly to obtain the equation of the normal. It was evident that some learners had used a calculator to find the value for the derivative and had not shown all the stages of their working. Some learners who did find a correct equation of the normal were either unable to rearrange their final equation accurately or they did not write it in the required format.

Part (b)

In part (b), many learners either successfully obtained the correct answer or correctly followed through their answer to part (a). Occasional transcription errors were seen, such as dividing by 4 instead of 14.

Part (c)

In part (c), finding the area of the trapezium proved problematic for many learners. A common error was to find the area of one triangle, usually the one encompassing the trapezium, but not the smaller triangle that needed to be deducted. Learners frequently made mistakes evaluating these triangle areas by mixing up the x and y coordinates. Some used integration to find the trapezium area. When attempted, this was generally done well, but some used the incorrect limits, usually choosing limits that found the area of a larger triangle.

For the integration to find the area under the curve, those who spotted the reverse chain rule completed this quickly and easily. Attempts using substitution were the next most successful but usually took far more work and many responses showed errors in the processing e.g. the limits were not always

changed. Some learners attempted to use integration by parts but were usually unsuccessful in attaining the required form. A correct value for the area under the curve coming from an incorrect integration was often seen with learners having used calculators to evaluate the area.

Question 11

This question tested modelling using trigonometry. Learners were able to make a good start in many cases, however, parts (b) and (c) were rarely both answered correctly; often responses showed some progress in each, but manipulation errors or a misunderstanding of what was being asked compromised the marks scored. It was pleasing to see nearly all learners attempting the final part which required them to engage with the model and consider the appropriateness of it.

Part (a)

Part (a) was answered extremely well, with the vast majority of learners successfully expressing the trigonometric expression in the required harmonic form. Most learners correctly found the exact value of R together with an appropriate value of α . Common errors included giving R as a decimal rather than an exact value, giving α to three significant figures instead of three decimal places, or obtaining a negative value of α . Although many learners demonstrated a sound understanding of the technique, some were less successful in applying their result to the later parts of the question.

Part (b)

Learners who used the result from part (a) generally found part (b) straightforward and were able to identify the minimum demand and determine the corresponding time of day. Common errors arose when learners attempted to work directly with the original expression, often treating the sine and cosine terms independently by assuming both could simultaneously attain their

minimum values. Some learners identified the maximum rather than the minimum demand, set the cosine term equal to 0 instead of -1 , or omitted the phase shift when solving for the time. Most learners who obtained a correct decimal value for the time successfully converted this into hours and minutes, although a few interpreted the decimal part as minutes. Learners should also be reminded to state times unambiguously using an appropriate 24-hour time or am/pm notation, as the question was dependent on identifying the correct time of day.

Part (c)

Part (c) proved more demanding. Whilst many learners recognised that differentiation was required, a significant number simply substituted $t = 15$ into the original model. Of those who used differentiation, the most common error was omission of the chain rule factor $\frac{\pi}{12}$, although some learners attempted to use its reciprocal or introduced unnecessary complexity by first expanding the trigonometric expression. A further error was to substitute $t = 3$ instead of $t = 15$ not recognising that t represented the number of hours after midnight. Learners were equally likely to differentiate either the harmonic form obtained using part (a) or the original model, with both approaches often leading to fully correct solutions when carried out accurately. Some responses just stated a value given this could be done on a calculator; these did not score any marks.

Part (d)

Part (d) was answered correctly by almost all learners. Most recognised that electricity demand varies throughout the year due to seasonal changes, although references to holidays, weather or other special events affecting demand were also commonly seen and accepted.

Question 12

Drawing a sketch of the modulus graph in part (a) was generally answered well including learners who found the paper more challenging. Part (b) involving the intersection of the modulus graph with a reciprocal graph proved to be one of the more demanding questions on the paper.

Part (a)

Most learners sketched a “V” shape graph showing their understanding of the generic shape of a modulus graph. Many responses had correctly positioned graphs and these learners often correctly labelled the points of intersection with the coordinate axes. Weaker responses appeared not to have taken the “ q ” into consideration when determining the coordinates of the vertex. Learners could find these coordinates by setting $y = 0$ and solving the resulting $px - q = 0$, and this method usually resulted in the correct answer. The coordinates of the y -axis intersection were usually given correctly although some learners were unable to achieve this mark as their graph stopped at the y -axis instead of passing through it. Some learners labelled the positive y -intercept as $-q$, seemingly not realising that this was impossible as q was a positive constant. Centres may wish to encourage learners to look out for any conditions placed on constants, particularly if there is emboldened writing in the question.

Some learners sketched graphs that were not V shaped, although the mark for the intersection with the y -axis was still available. A few learners appeared to take advantage of their graphical calculators and sketched a graph with p and q replaced by numerical constants to work out the shape of the graph. Those that went on to consider the coordinates of the intersections with the axes were able to secure full marks, but weaker responses left the coordinates in terms of the chosen values so were unable to score further marks.

Part (b)

Many responses did not demonstrate a good understanding of this part of the question and often no attempt was seen. Stronger responses saw the graph of $y = \frac{k}{x}$ added to the answer to part (a) which may well have helped the learner to work out a correct strategy. Many learners set the equations of the two graphs equal to each other, but often only considered the right-hand branch of the modulus graph $\left(px - q = \frac{k}{x}\right)$ which would not lead to a correct solution. Having created either one or two equations, weaker responses made no further progress so no marks could be awarded. In stronger responses, the left-hand branch was considered $\left(-px + q = \frac{k}{x}\right)$, the resulting equation was manipulated correctly to achieve a three-term quadratic equation and $b^2 - 4ac > 0$ was correctly applied. Those learners who had only considered the right-hand branch of the modulus graph were still able to score the second method mark for applying $b^2 - 4ac > 0$. Sometimes the discriminant was applied to the equations formed from considering both branches; better responses then correctly dismissed the possibility that k could be negative as the signs of all the constants were considered.

Some responses used other approaches such as the differentiation of $y = \frac{k}{x}$ and comparing gradients whilst completing the square was also seen occasionally; both these methods had varying degrees of success.

Question 13

This question was one of the most demanding on the paper; it required learners to find the common ratio and sum of a geometric sequence where the terms were expressed as trigonometric ratios.

Almost all learners attempted and made some progress on this question.

The calculator warning stated that one should not be used at all for this question, although it was clear in many cases that one had been used where inexact values were given as decimals. In such cases responses did not score the marks. However, it was condoned if not as many stages of working was shown for this question provided the solution itself would not have required a calculator to be used e.g. not showing the manipulation when multiplying fractions involving surds was not penalised nor the use of the summation formula without rationalising the denominator $1 - \sqrt{6}$.

Part (a)

In part (a), overall, a solid grasp of how to initiate geometric sequence problems by equating terms for the common ratio was demonstrated.

After using the three terms correctly to set up an equation in θ , a variety of different routes through rearrangements were taken to achieve either $\cos \theta = \frac{\sqrt{3}}{2}$

, $\tan \theta = \frac{1}{\sqrt{3}}$ or $\operatorname{cosec} \theta = 2$ and $\theta = \frac{\pi}{6}$, followed by a correct attempt to achieve $r = \sqrt{6}$.

As the manipulation to achieve a solvable equation required using trigonometric identities, some slips were made when dealing with fractions within fractions, and the most efficient route of using $\cot \theta = \frac{\cos \theta}{\sin \theta}$, cross-multiplying and achieving

$2 \cos^2 \theta = \sqrt{3} \cos \theta$ leading to $\cos \theta = \frac{\sqrt{3}}{2}$ was not the obvious route for a number of learners. A minority of responses were seen where the equation was set up using

$\frac{u_2}{u_3} = \frac{u_1}{u_2}$ which led to a correct value for $\cos \theta$ but then a value for $\frac{1}{r}$ which was

considered as r in subsequent work. Some learners did not proceed to find r using their value for θ in part (a), but if it was subsequently seen as correct in part (b), they achieved the A mark. A misconception was to consider $\frac{\pi}{6}$ as r . The responses which started off with $ar^2 = \sin \theta$ and $ar^3 = \sqrt{2} \cos \theta$ were less successful, often not achieving a solvable equation in θ .

Part (b)

In part (b) most learners recognised that they needed to find a value for a by dividing u_3 by r^2 . It was quite common to see $6a = \frac{1}{2}$ leading to $a = 3$ rather than

$a = \frac{1}{12}$. The sum of the first three terms was found either by adding $\frac{1}{12}$, $\frac{\sqrt{6}}{12}$ and

$\frac{6}{12}$ or using the sum formula. Some responses showed an incorrect use of $\frac{\pi}{6}$ as

r . A very common response was finding the sum of the three terms given in the question. This was an alternative method, which if followed by dividing their sum by r^2 , would give the correct answer. However it was extremely rare to see this,

so these responses only achieved one mark for $\frac{1}{2} + \frac{\sqrt{6}}{2} + 3$ or using the sum formula.

Question 14

This question assessed learners' ability to solve a differential equation by separation of variables, and to apply the resulting model to answer questions involving a maximum value and a specific function value. Part (a) required learners to separate variables, integrate, and use the given conditions to determine unknown constants, before rearranging into the required form. Parts (b) and (c)

required learners to apply their model, including differentiation to establish a maximum and to solve for a given value of the model.

Most learners were able to secure the first two method marks in part (a), correctly separating the variables and using the given conditions to determine their constants; however, many weaker responses which involved rearranging the equation into the required form using logarithm and index laws were undermined by inconsistent algebraic manipulation.

Part (a)

Learners generally separated the variables correctly and integrated to reach an expression of the form $\frac{1}{2} \ln x = At - \frac{t^2}{2} + c$, or an equivalent, before successfully using the given conditions to determine their constants. Responses that clearly identified which condition had been used to find each constant tended to progress most smoothly through this stage. Some learners chose to find integrating factor to integrate, which was generally less successful than the standard method.

Difficulties tended to arise in the final rearrangement, where learners were required to manipulate their equation into the form $x = f(t)$. Common errors included combining logarithmic terms, inconsistent handling of the $\ln 0.3$ term, or sign errors when moving from logarithmic to exponential form. Learners are encouraged to take particular care with the order of operations at this stage, and to ensure their final expression is presented explicitly as x in terms of t .

Part (b)

This part was answered less successfully overall, with relatively few learners deducing that $t = A$ corresponded to the time at which the maximum concentration occurs. Where this was not identified, the differentiation

techniques used further limited learners' ability to access the marks available in this part of the question.

Part (c)

Learners who substituted $x = 0.1$ into their equation from part (a) generally obtained an equation involving an exponential term in t , which could then be rearranged into a three-term quadratic. Responses that clearly set out this rearrangement before attempting to solve tended to be more successful, with the quadratic solved by any correct method, including the use of a calculator. Where learners did not gain marks, this was most often due to errors in forming the three-term quadratic from their exponential equation, or through jumping directly to a final value for T without showing sufficient intermediate working to support the method mark available.

Question 15

This question focused on proof by contradiction and required learners to show detailed algebraic reasoning. It appeared to be particularly challenging for many learners.

The starting point provided by the question enabled most learners to start the question by substituting the given algebraic expressions for distinct odd numbers into $a^2 + b^2$ and expanding the brackets to achieve the first method mark. After this, many were unsure how to proceed and scored no further marks.

Some responses did not simplify their terms; of those who went on to collect terms or factorise to reach $4m^2 + 4m + 4n^2 + 4n + 2$ or $2(2m^2 + 2m + 2n^2 + 2n + 1)$ some stopped without explicitly stating that their expression was even or deducing that c^2 was even and so failed to score the first accuracy mark. Some learners attempted an alternative approach of explaining the contradiction at this

stage purely in words (such as arguing that odd + odd = even), but this did not provide the algebraic reasoning required by the question.

Many learners who had made it as far as stating their expression was even or deducing that c^2 was even, were not able to make sufficient progress with some stating a final conclusion at this point, by incorrectly assuming they had completed the proof. Other responses did not have an equation which would lead to a valid contradiction. For example, they did not have an equation set up using $c = 2k$ after stating c was even in order to reach $c^2 = 4k^2$. Some responses did not deduce that c was even and instead considered both cases of c being odd or even; this could still lead to a fully correct proof.

Common misconceptions included attempting to take the square root of the expanded $a^2 + b^2$ expression and falsely claiming a contradiction based on arguments about irrationality that received no further credit. This approach frequently resulted in erroneous algebraic operations, such as attempting to evaluate $\sqrt{2m^2 + 2m + 2n^2 + 2n + 1}$ by incorrectly taking the square root of each individual term. Some learners incorrectly set $c^2 = 2k^2$ which led to an invalid deduction. Of the responses which had successfully arrived at $c^2 = 4k^2$, many showed no further progress as they did not follow up to set up a full argument leading to a contradiction such as dividing both sides of their equation by 2 or attempting to write both sides as multiples of 4.

Some learners did not achieve the final mark even though they had achieved the previous three marks because they did not include a conclusion. Of those who were successful most took the approach of dividing both sides of the equation by 2 to create one side which was even and one side which was odd, directly leading to the required mathematical contradiction. Although not seen as often some learners instead considered multiples of 4 by taking out a factor of 4 from both sides or dividing each side of the equation by 4.